

Deep Learning for Structured Data – Exercise Sheet 2

Submit the solution in a single PDF in the Moodle until 15.05.2026, 12:00. The corresponding exercise session will take place on 22.05.2026 at 08:30.

Problem 1: (2 Pt.)

For each of the following maps, decide whether it is invariant, equivariant, or neither under permutations of the entries of $x \in \mathbb{R}^m$:

1. $f(x) = \sum_{i=1}^m x_i$,
2. $f(x) = \max_{i \in \{1, \dots, m\}} x_i$,
3. $f(x) = x$,
4. $f(x) = (x_1 + \dots + x_m)\mathbf{1}$,
5. $f(x) = (x_1, x_1, \dots, x_1)$.

Justify your answer with formal proofs.

Problem 2: (1 Pt.)

Let $f: \mathbb{R}^{m \times d} \rightarrow \mathbb{R}^{m \times d'}$ and $g: \mathbb{R}^{m \times d'} \rightarrow \mathbb{R}^{m \times d''}$ be S_m -equivariant maps, and let $\rho: \mathbb{R}^{m \times d''} \rightarrow \mathbb{R}^k$ be S_m -invariant.

1. Prove that $g \circ f$ is S_m -equivariant.
2. Prove that $\rho \circ g \circ f$ is S_m -invariant.

Problem 3: (1 Pt.)

Let $u: \mathbb{R}^{m \times d} \rightarrow \mathbb{R}$ be permutation-invariant. Define the map $F: \mathbb{R}^{m \times d} \rightarrow \mathbb{R}^m$ by

$$F(X)_i := u(X) \cdot u(X_{-i}), \quad i \in [m],$$

where X_{-i} denotes the collection obtained from X by removing its i -th element. Prove that F is permutation-equivariant.

Problem 4: (2 Pt.)

In the lecture, we derived a precise characterization of all S_m -equivariant linear layers. Prove the corollary that extends the result to multi-dimensional linear mappings.

Problem 5: (2 Pt.)

Consider the DeepSets architecture from the lecture,

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right).$$

Replace the sum aggregation with each of the following alternatives:

1. mean,
2. max,
3. point-wise multiplication.

For each choice, determine whether the resulting architecture remains permutation-invariant via a formal proof. In addition, for each of the above aggregation functions, find out what information about the input set may be lost. That is, derive a function that can be expressed using sum aggregation but not by the above aggregation function.

Problem 6: (1 Pt.)

The slides define momentum updates by

$$\mathbf{g}_t = \nabla_{\boldsymbol{\theta}_t} F(\boldsymbol{\theta}_t), \quad \mathbf{v}_{t+1} = \beta \mathbf{v}_t + \mathbf{g}_t, \quad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \mathbf{v}_{t+1},$$

with $\mathbf{v}_0 = \mathbf{0}$.

1. Show by induction that

$$\mathbf{v}_{t+1} = \sum_{k=0}^t \beta^{t-k} \mathbf{g}_k.$$

2. Explain why this is called an exponentially weighted moving average.
3. What happens when $\beta = 0$?

Problem 7: (2 Pt.)

Design a non-permutation-equivariant architecture on sets and devise a data distribution on sets. Show that there exists a training dataset so that the DeepSets architecture, for some parameter assignment, achieves a perfect test score on arbitrary data sampled from the constructed distribution, while the non-permutation-equivariant architecture's prediction is arbitrarily bad.

Problem 8: (3 Pt.)

Make yourself familiar with PyTorch Geometric, e.g., by following the tutorial at <https://pytorch-geometric.readthedocs.io/en/latest/>. Then,

1. implement the PointNet architecture,
2. evaluate your implementation against a standard DeepSet architecture on a point cloud dataset of your choice from <https://pytorch-geometric.readthedocs.io/en/2.5.1/modules/datasets.html>.

Problem 9: (2 Pt.)

Implement and compare three aggregation functions in a DeepSets architecture: sum, mean, and max.

1. Implement three models of the form

$$f(X) = \rho(\text{agg}_{x \in X} \phi(x)),$$

where $\text{agg} \in \{\text{sum}, \text{mean}, \text{max}\}$.

2. Construct a synthetic task where the label depends on the cardinality of the set, and explain which aggregation functions succeed or fail.